Finance1000 Visual Data Analysis

Figure-first report for a 1000-ticker hourly US equity panel.

Universe: 1000 tickers

Timestamps: 28,510 hourly rows

Span: 2015-01-02 00:00:00+00:00 to 2026-02-25 00:00:00+00:00

Representative tickers: AAPL, MSFT, AMZN, JPM, KMB, NVAX, GME, META, FCEL, MARA, CL, VZ

How to read this report

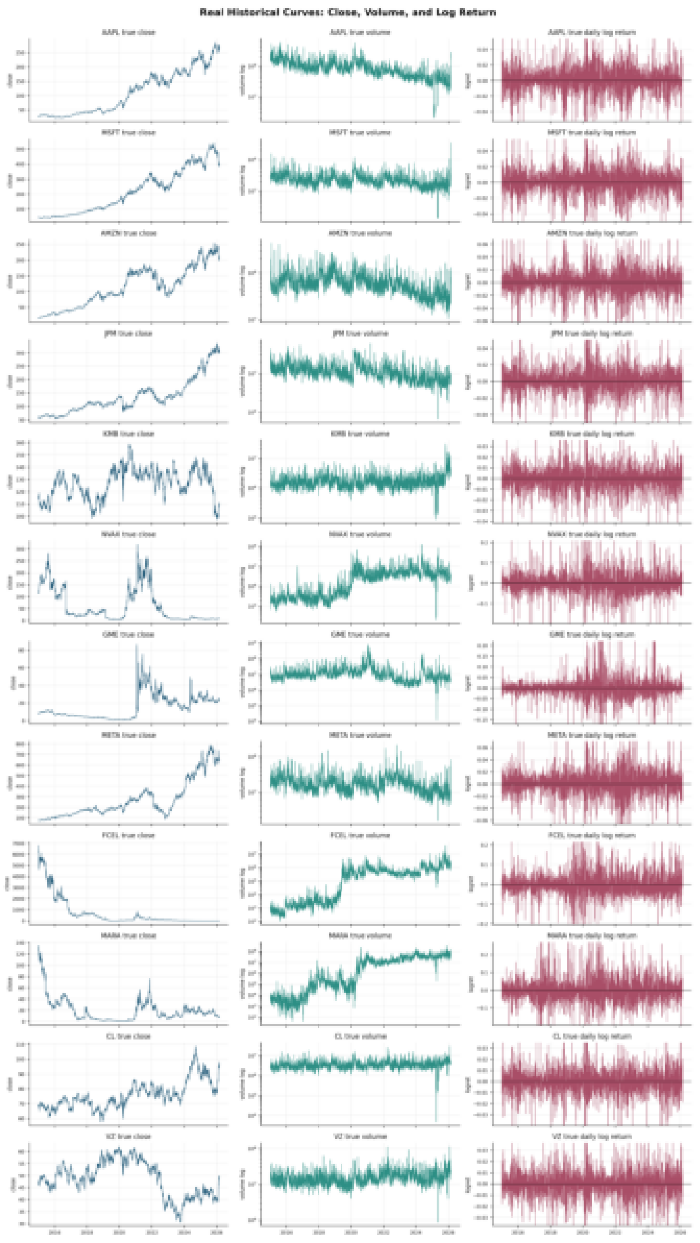
- Each section is organized around one generated figure and a short interpretation.
- The goal is visual data validation: understand raw scale, transformed log-return behavior, tails, cross-sectional structure, and ticker heterogeneity before modeling.
- The separate ticker book contains one page per ticker for detailed inspection of all 1000 series.

Distribution Statistics Snapshot

group	count	mean	std	p1	p50	p99
Raw close	7,000,000	2.56742e+07	2.16032e+09	0	33.06	15300
Raw volume	7,000,000	363623	1.27387e+06	0	89412	4.32553e+06
Close log return	7,000,000	1.47449e-05	0.0169491	-0.0373178	0	0.0381934
Volume log change	7,000,000	0.00354011	1.32898	-2.16791	0	2.27764

Interpretation: raw close and volume are highly skewed, while log returns are centered near zero but remain heavy-tailed. These statistics motivate reporting both raw-data views and log-return diagnostics.

1. Real curves: close, volume, and log return



Why this figure matters

This is the first sanity check: the report shows actual historical trajectories rather than synthetic examples.

Reading notes

- Each row is one representative ticker; columns are daily close, daily volume, and daily close log return.

- The close panels reveal regime shifts, split-like artifacts, outliers, and long-run trends that affect model scaling.

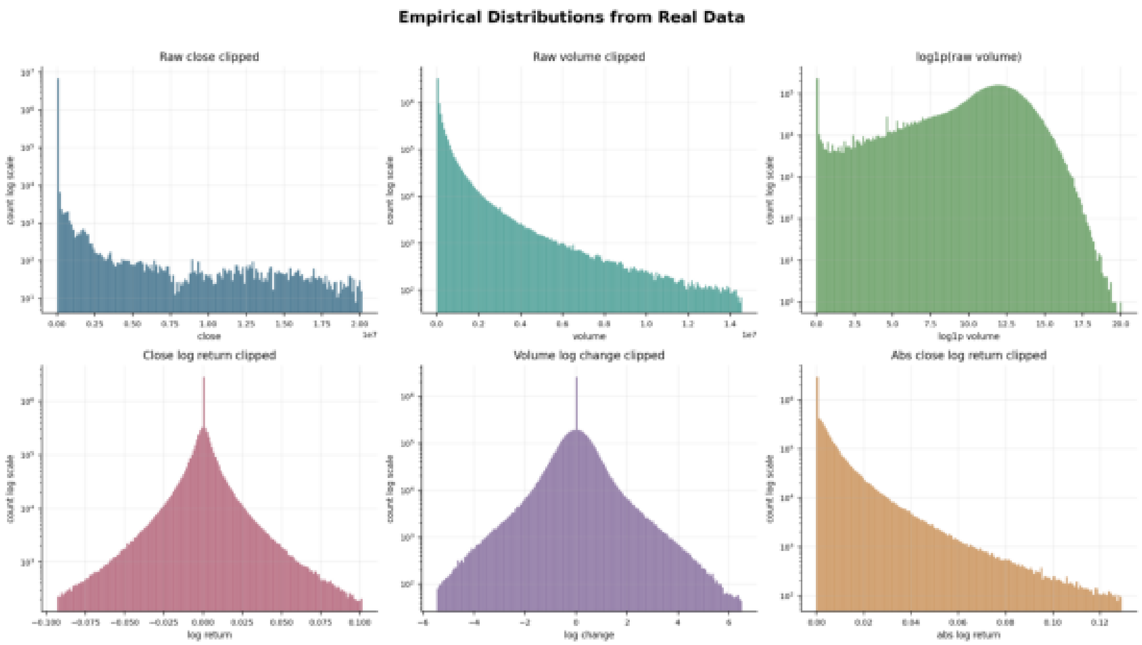
- The volume panels use a log scale because liquidity is extremely uneven across tickers and time.

- The log-return panels make volatility clustering and jump periods visible in a stationary-like representation.

Why this figure matters

Raw prices and volumes are heavy-tailed; log-return views are needed for model diagnostics and robust comparison.

2. Empirical distributions: raw and transformed data



Reading notes

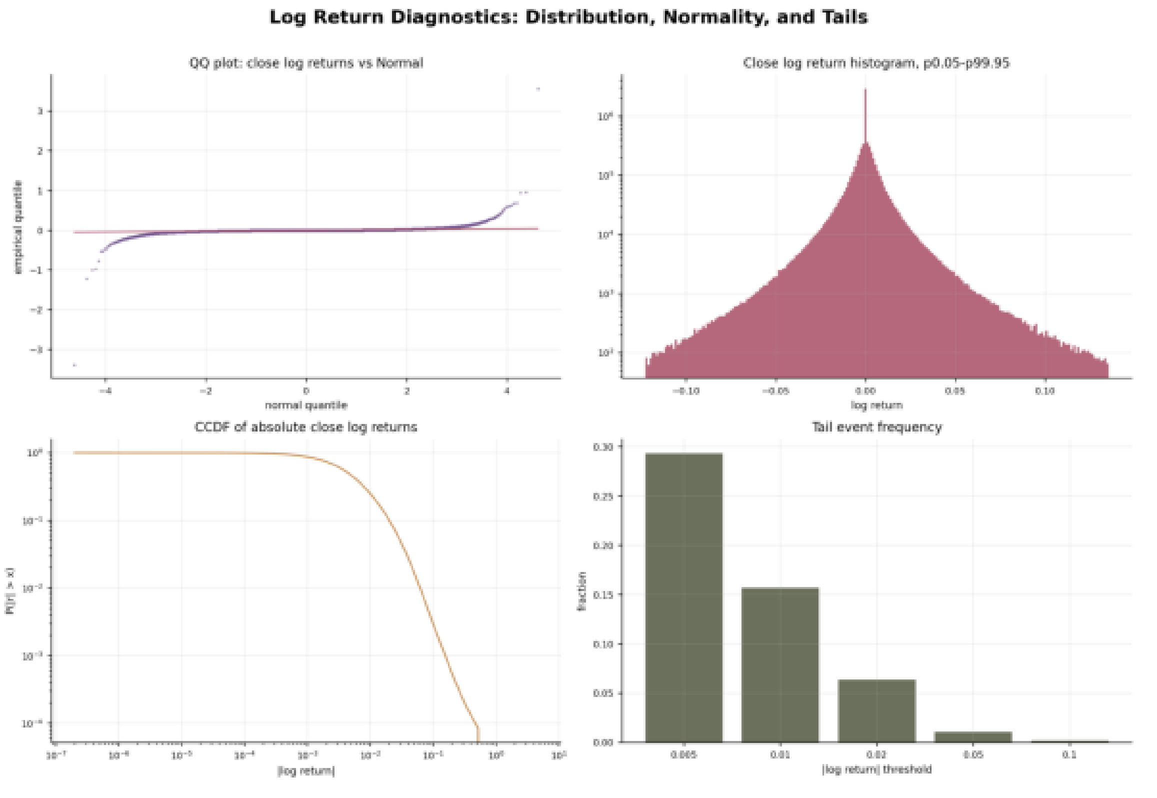
- Raw close and raw volume distributions are clipped at extreme quantiles for readability while preserving log-count scaling.

- The log1p(volume) panel compresses liquidity differences while preserving zero-volume structure.

- Close log returns are centered near zero but have heavy tails, which is typical for equity data.

- Absolute log returns expose the tail mass that drives forecasting error and risk-sensitive metrics.

3. Log-return diagnostics: normality and tails



Why this figure matters

This page checks whether the transformed series behaves like a simple Gaussian process. It does not.

Reading notes

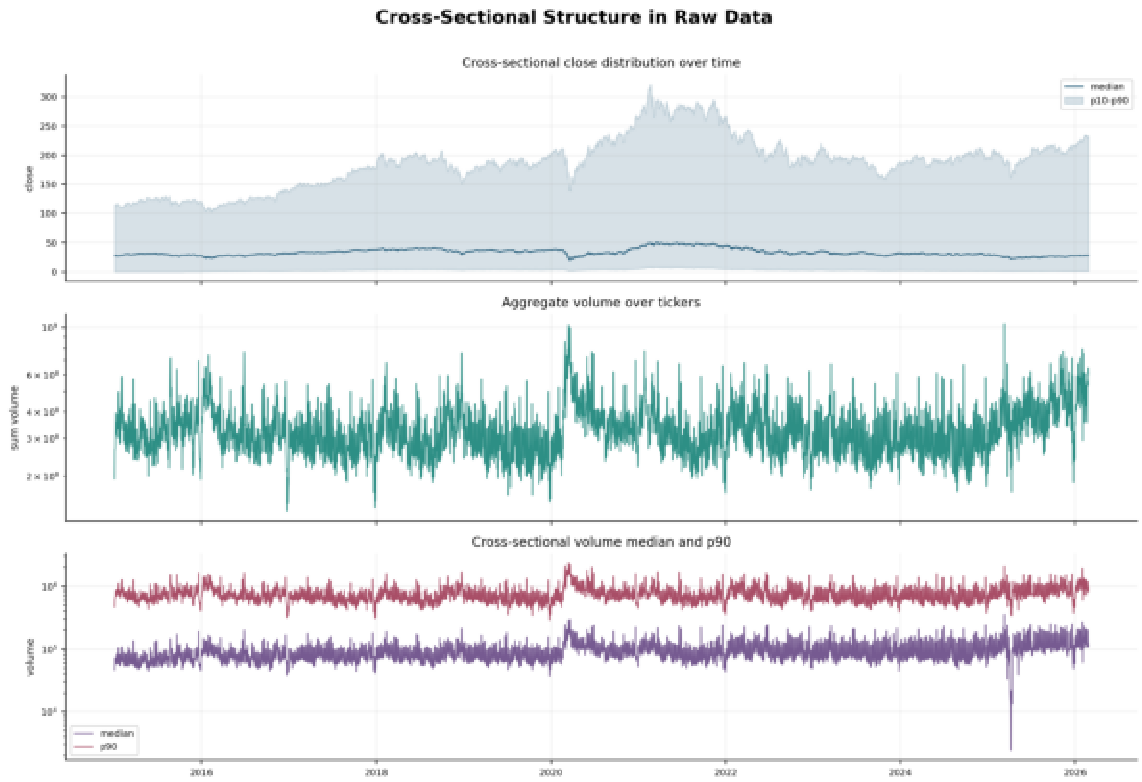
- The QQ plot compares empirical close log returns with Normal quantiles; tail deviations indicate non-Gaussian behavior.

- The clipped histogram shows the central mass while preserving log-scale count differences.

- The absolute-return CCDF highlights tail decay and rare but material jumps.

- Tail event frequencies quantify how often moves exceed fixed log-return thresholds.

4. Cross-sectional structure over time



Why this figure matters

Foundation models see a panel, not one isolated ticker; cross-sectional scale changes matter for normalization and sampling.

Reading notes

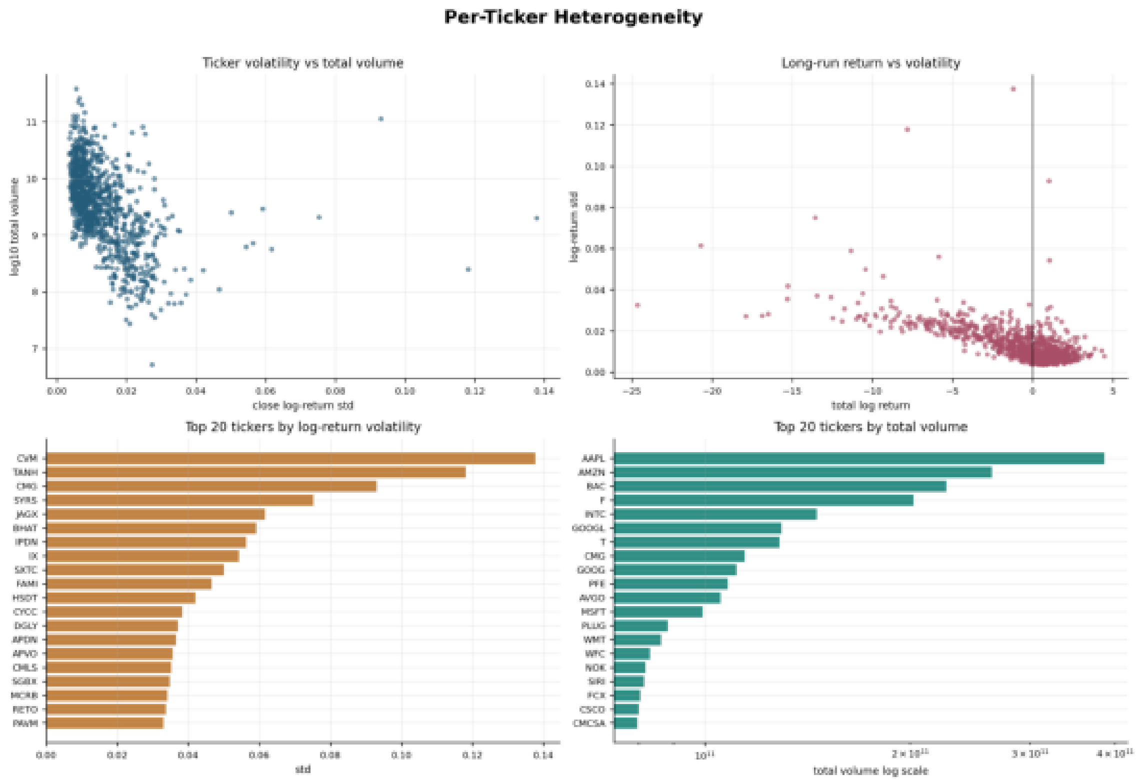
- The close p10/p50/p90 band tracks how the price universe changes over calendar time.

- Aggregate volume shows market-wide liquidity regimes and calendar effects.

- Volume median and p90 show that liquidity is concentrated in a subset of tickers.

- These plots are useful for detecting broad data shifts before model evaluation.

5. Per-ticker heterogeneity



Why this figure matters

A 1000-ticker dataset mixes liquid mega-caps, quiet names, and highly volatile small caps; model errors will not be uniform.

Reading notes

- Volatility versus total volume separates liquid stable names from thin or unstable names.

- Long-run return versus volatility identifies tickers with high realized trend and high uncertainty.

- Top-volatility and top-volume rankings provide concrete candidates for stress tests and qualitative inspection.

- This page helps choose representative subsets for demos and deeper error analysis.

6. Heatmaps: return and rolling volatility regimes

Why this figure matters

Heatmaps make simultaneous time structure visible across representative and high-volatility tickers.

Reading notes

- The daily log-return heatmap shows co-moving shock periods and ticker-specific jumps.

- The rolling annualized volatility heatmap shows persistent risk regimes rather than isolated spikes.

- Representative tickers are mixed with high-volatility names to expose both normal and stressed behavior.

- This view is useful for selecting forecast windows for qualitative foundation-model comparisons.